

Übung Galerkin:

$$-u'' + u = x \quad \text{auf } [0,1] \quad u(0)=0 \text{ und } u(1)=0$$

1. Schwache Lösung bilden

$$(-u'' + u)v = xv, \text{ mit } v \in C_c^\infty$$

$$\int_0^1 (-u'' + u)v dx = \int_0^1 xv dx$$

$$-\int_0^1 u''v dx + \int_0^1 uv dx = \int_0^1 xv dx$$

$$\int_0^1 u'v' dx + \int_0^1 uv dx = \int_0^1 xv dx$$

$$\stackrel{!}{=} a(u,v) = L(v)$$

2. $\varphi_1 = x(1-x)$

$$u_a = c_1 \cdot \varphi_1(x)$$

$$u_a = c_1 \cdot x(1-x)$$

$$\varphi_1'(x) = 1 - 2x$$

$$(x - x^2)^2 = x^2 - 2x^3 + x^4$$

3. Linke Seite $(a(u, \varphi))$ lösen

$$a(u, \varphi) = \int_0^1 c_1(1-2x) \cdot (1-2x) dx + \int_0^1 c_1 x(1-x) x(1-x) dx$$

$$= \int_0^1 c_1(4x^2 - 4x + 1) dx + \int_0^1 c_1(x^4 - 2x^3 + x^2) dx$$

$$= \left[c_1 \left(\frac{4}{3}x^3 - 2x^2 + x \right) \right]_0^1 + \left[c_1 \left(\frac{1}{5}x^5 - \frac{1}{2}x^4 + \frac{1}{3}x^3 \right) \right]_0^1$$

$$= \frac{1}{3}c_1 + c_1 \left(\frac{6}{30} - \frac{15}{30} + \frac{10}{30} \right)$$

$$= \frac{1}{3}c_1 + \frac{1}{30}c_1$$

$$= \frac{11}{30}c_1$$

4. Rechte Seite

$$L(\varphi_1) = \int_0^1 x x(1-x) dx = \int_0^1 x^2 - x^3 dx = \left[\frac{1}{3}x^3 - \frac{1}{4}x^4 \right]_0^1 = \frac{1}{12}$$

5. Zusammen führen

$$a(u, \varphi_1) = L(\varphi_1)$$

$$\frac{11}{30}c_1 = \frac{1}{12}$$

$$c_1 = \frac{5}{22} \Rightarrow \tilde{u}(x) = \frac{5}{22}x(1-x)$$

